

1) A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

Interpretation of the Reward Trend

The gradual increase in reward from 10 to 45 indicates that the agent is learning from its interactions with the environment. In the beginning, the agent receives a low reward because its policy is not yet effective. As the number of episodes increases, the agent learns which actions produce better outcomes.

Learning Performance

The increasing reward indicates improved learning performance. A higher reward means the agent is making better decisions and achieving more desirable results in the environment.

Policy Improvement

The trend also indicates policy improvement. The agent gradually updates its policy based on the rewards received. Therefore, it becomes more capable of selecting actions that maximize future rewards.

Conclusion

The increase in rewards from 10 to 45 shows that the RL agent is successfully learning and improving its policy. The upward trend demonstrates that the agent's decisions are becoming more effective over time. If the reward eventually becomes stable at a high value, it may indicate that the agent's policy is approaching convergence.

2) An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

Interpretation

$\epsilon = 0.1$:

The agent explores only about 10% of the time and exploits its learned knowledge about 90% of the time. Since the agent has already learned useful actions, exploiting them more frequently results in a higher cumulative reward.

$\epsilon = 0.3$:

The agent explores more frequently, approximately 30% of the time. This allows it to discover new actions but also causes it to sometimes choose actions that give lower rewards. Therefore, the cumulative reward is lower than with $\epsilon = 0.1$.

$\epsilon = 0.5$:

The agent explores approximately 50% of the time. It therefore makes random choices more frequently instead of selecting the best-known action. This results in a lower cumulative reward.

Best Epsilon Value

The best-performing value is:

$\epsilon = 0.1$

because it produces the highest cumulative reward. It provides a good balance where the agent mostly exploits the learned policy while still performing some exploration.

Conclusion

As epsilon increases from $0.1 \rightarrow 0.3 \rightarrow 0.5$, exploration increases and exploitation decreases. In this experiment, excessive exploration reduces the cumulative reward. Therefore, $\epsilon = 0.1$ provides the best performance among the tested values.

3) Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

The two Reinforcement Learning algorithms being compared are Q-Learning and SARSA based on their average rewards over 100, 200, and 300 episodes.

Performance Difference

- Q-Learning consistently achieves slightly higher average rewards than SARSA at 100, 200, and 300 episodes.
- This means Q-Learning is learning a more rewarding policy under the given experimental conditions.
- SARSA also improves with more episodes, but its reward remains slightly below Q-Learning.

Convergence Comparison

Since Q-Learning produces higher rewards consistently, it reaches better performance more quickly in the observed training period.

Therefore:

Q-Learning → Faster convergence and better performance

SARSA → Slightly slower convergence and lower average reward

Why?

Q-Learning is an off-policy algorithm. It learns the value of the best possible action, even if the agent currently follows a different action because of exploration.

SARSA is an on-policy algorithm. It learns based on the action that the agent actually takes, including exploratory actions. This can make SARSA more conservative and sometimes result in slightly lower rewards.

Conclusion

Based on the observed results over 100, 200, and 300 episodes, Q-Learning performs better and converges faster than SARSA, because it consistently obtains higher average rewards. However, this conclusion is valid for the given experiment; strictly proving convergence speed would require examining when each algorithm reaches a stable reward level.

4) An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

The **discount factor (γ)** determines how much the RL agent values **future rewards** compared with **immediate rewards**.

- **$\gamma = 0.2$:** The agent gives more importance to immediate rewards and has less concern for future rewards. It mainly focuses on short-term decisions.
- **$\gamma = 0.5$:** The agent considers both immediate and future rewards with moderate importance.
- **$\gamma = 0.9$:** The agent gives high importance to future rewards and therefore focuses more on long-term benefits.

Effect on Long-Term Reward Optimization

As the discount factor increases from **0.2** $\rightarrow$ **0.9**, the agent considers future rewards more strongly. This helps the agent learn actions that may provide smaller rewards initially but produce **higher rewards in the long run**.

Since the question states that the **average reward increases as the discount factor increases**, the agent performs better when it considers future rewards.

Comparison

Discount Factor	Focus	Learning Performance
0.2	Short-term rewards	LOW
0.9	Long-term rewards	HIGH

Most Suitable Value

$\gamma = 0.9$ is the most suitable among the given values because it produces the **highest average reward**. It allows the agent to consider future consequences of its actions and optimize its policy for better long-term performance.

Conclusion

The increase in average reward from $\gamma = 0.2$ to $\gamma = 0.9$ shows that considering future rewards improves the agent's learning performance in this experiment. Therefore, $\gamma = 0.9$ **provides the best performance** among the tested discount factors.

5) The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

The success rate of the Reinforcement Learning (RL) agent increases from 45% at 50 episodes to 92% at 200 episodes.

Improvement in Success Rate

- At 50 episodes, the success rate is 45%, indicating that the agent is still learning and makes many incorrect decisions.
- At 200 episodes, the success rate reaches 92%, showing significant improvement in the agent's ability to choose successful actions.
- The success rate increased by 47 percentage points, which indicates strong learning progress.

Interpretation

The increasing success rate indicates that the agent is learning a better policy as the number of training episodes increases. The agent is gradually understanding which actions lead to successful outcomes.

A success rate of 92% is very high, suggesting that the learned policy is close to optimal for the given environment.

However, we cannot definitively say that the policy is optimal only from these two values. To confirm optimality, we would need to observe whether:

- The success rate remains consistently high over additional episodes.
- The performance has converged or stabilized.
- The agent performs well on unseen/test episodes.
- There is little variation in performance.

Conclusion

The increase from 45% to 92% demonstrates significant improvement in learning performance and policy quality. The agent has learned an effective policy and may be approaching an optimal policy, but additional training and evaluation are required to confirm that it has actually reached the optimum.

